islandOS

AI-Powered Forecasting and Multi-Resource Coordination for Energy-Constrained Infrastructure

A simulation-first decision-support platform that helps energy-constrained sites anticipate changing energy conditions, coordinate distributed resources, and compare operating strategies through transparent, cost-aware analysis.

PROGRAMME	Global Sustainability Challenge 2026–27 · Phase 1 — IGNITE
THEME	Sustainable Energy
PROJECT STAGE	Early-stage software prototype

Better energy decisions begin before the next operating hour.

From fragmented operating data to forecast-informed, constraint-aware and economically transparent decisions.

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REPOSITORY	https://github.com/HaochuanLin/islandOS
LIVE DEMO	https://islandos.streamlit.app/

> All built-in demonstration inputs are synthetic. IslandOS is an early-stage decision-support prototype. No field-validated accuracy, verified customer savings or commercial deployment is claimed.

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> **HOW TO READ STATUS CLAIMS IN THIS DOCUMENT**
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> Every technical claim carries one of four labels. A label describes the state of the software evidence, not the ambition of the project. No claim in this document is field-validated.

Label	Definition used throughout this proposal
IMPLEMENTED	A capability that runs on the default branch within the explicitly stated scope.
EXPERIMENTAL	A working exploratory implementation or limited workflow that has not been fully integrated into the main customer-facing workflow.
PLANNED	A designed capability that is not yet available as a usable workflow.
LONG-TERM	A future capability dependent on site integration, engineering approval, cybersecurity review and explicit authorisation.

Passing software tests demonstrates software behaviour under tested conditions; it does not establish field performance or operational safety. A merged pull request is not field validation, and a hosted demonstration page is not a deployed customer system.

This proposal is written so that the distinction between built, experimental, planned and future capability is visible on every page. Where a statement depends on evidence the team does not yet hold, it is marked as a hypothesis, a design objective or an open validation question.

1. Executive Summary

Energy-constrained sites face a coordination problem, not simply an energy-generation problem.

Remote communities, weak-grid facilities and distributed infrastructure may depend on a changing mix of solar and wind generation, batteries, grid electricity, diesel backup and electricity-consuming services. These resources have different costs, operating limits and availability patterns. Their supply and demand conditions can change throughout the day. Operators must therefore make interconnected decisions: how much energy to reserve, when to use storage, when to rely on backup generation, which flexible loads can be shifted, and how to maintain essential services without incurring unnecessary operating costs.

islandOS is an AI-assisted, simulation-first decision-support platform designed to help address this challenge. It brings together site data, forecasting methods, energy-system simulation, constrained resource coordination, scenario comparison and customer-economics screening in a modular software architecture. The intended workflow is straightforward:

01	02	03	04	05
INPUT	**FORECAST**	**COORDINATE**	**COMPARE**	**DECIDE**
Organise operating data, equipment information, tariffs and constraints.	Anticipate changes in energy supply, demand and relevant operating signals.	Evaluate how available resources and flexible loads may operate together.	Assess alternative strategies under consistent physical and economic assumptions.	Present understandable trade-offs and auditable information for human review.

The central value proposition is not autonomous control or a claim of superior prediction accuracy. It is the ability to turn fragmented operating information into a more structured, explainable and economically assessable decision process.

The current project is an early-stage software prototype. It already contains data-onboarding foundations, simplified energy-system simulation, strategy comparison, audit and provenance functions, customer-economics screening, and experimental forecasting and scheduling workflows. Some of these capabilities remain separate from the hosted end-to-end demonstration. The team has not yet completed a real-site pilot, obtained a representative operational dataset, established site-specific forecast accuracy or verified customer savings. The next stage is therefore to validate the platform with authorised site data, qualified engineering review and operator feedback. If validated, islandOS could develop into a reusable decision-support layer for microgrids and other distributed energy infrastructure, with future potential for integration with existing energy-management and equipment systems.

THE CUSTOMER VALUE

islandOS is designed to help an operator answer three practical questions:

FORESIGHT	What is likely to happen to energy supply and demand in the next operating period?
COMPARISON	Which feasible operating strategy offers the most appropriate balance of cost, reliability and resource use?
ACCOUNTABILITY	Can the operator understand, review and reproduce why a particular strategy was recommended?

5	5	4	1
STUDENT TEAM MEMBERS	**WORKFLOW STAGES**	**CORE RESOURCE GROUPS**	**NEXT MILESTONE**
four disciplines, two countries	input to decide	generation, storage, grid, diesel	one authorised pilot dataset

Next milestone. Secure one suitable pilot partner and obtain authorised historical operating data. The team does not yet have a pilot site, and this document does not assume one.

2. Problem Statement

2.1 Problem and context

Distributed energy systems in remote and weak-grid locations are rarely single-source. A typical hybrid site combines solar PV, sometimes wind, a battery bank, an import/export grid connection where one exists, one or more diesel generators, essential demand that must not be interrupted, and flexible demand that can be moved within a window — pumping, desalination, cold storage, grinding, or a shift-based industrial process.

Coordinating these assets is not a matter of following a fixed rule. Renewable output moves with weather. Demand moves with the day, the season and the water tank. Batteries have charge, discharge and state-of-charge limits. Generators have minimum loading and fuel constraints. Grid electricity may be time-priced, capped, or simply absent for hours at a time.



Figure 1. Energy–service coupling in a hybrid site. islandOS does not optimise one box in this chain; it represents the whole chain and surfaces the trade-offs between its links for human review.

COUPLED CONSTRAINTS

Weather variability · battery state-of-charge and cycle limits · diesel minimum load and fuel logistics · grid availability and tariffs · load flexibility windows · essential-service reliability · equipment ratings · operator approval before any set-point changes.

The challenge is particularly relevant where energy resources are distributed, operating conditions are variable, and specialist energy-modelling capacity is limited. A site may have several pieces of equipment that each operate correctly in isolation, yet their combined operation may not deliver the best overall outcome. For example, charging a battery, running a diesel generator, importing grid electricity and operating a flexible process are decisions that affect one another. A decision that appears economical for one asset may increase total system cost or reduce the reserve available for essential services. This creates a need for system-level decision support rather than isolated equipment monitoring.

2.2 The customer's decision problem

A site operator or budget owner has to answer concrete questions, usually under time pressure and without modelling support:

- What should operate, and when — battery, diesel backup, flexible load, or grid import?
- How should limited energy and water capacity be allocated between essential and flexible services?
- What should be reserved for critical service, and what is left for discretionary operation?
- What is the cost of each alternative, under the tariffs and fuel prices the site actually faces?
- What happens to essential services under each candidate strategy?
- Can the operator justify the decision to a board, a lender, a regulator or an off-taker?

THE CORE PROBLEM

Energy-constrained operators need to coordinate multiple resources under uncertainty, but often lack an accessible way to forecast changing conditions and compare the operational and economic consequences of alternative decisions.

The cost of a wrong decision is rarely a single bill. It appears as diesel burnt to cover a shortfall that a planned load shift would have avoided, as renewable output curtailed while a generator runs, as battery throughput spent on a peak that a tariff change would have made cheaper, and as service interruptions that erode willingness to pay. None of these costs is visible when each decision is made in isolation, which is why the comparison layer — not another dashboard — is the product.

This proposal does not claim that small energy sites have no energy-management software, nor that existing commercial products are incapable of these functions. The gap described here is a capacity-and-integration gap, not a claim of technical absence in the market.

2.3 Root causes · 2.4 Target users and stakeholders

Four root causes, three user roles

The problem is not a shortage of hardware. It resolves into four causes, each of which shapes the product:

Root cause	What it looks like on site	Design response in islandOS
Variable supply and demand	Decisions built on the current meter reading do not suit the next operating period; weather and load both move faster than the decision cycle.	A forecasting layer that produces an hourly outlook and states its uncertainty rather than a single number.
Interdependent equipment constraints	Batteries, generators, grid connections and flexible loads are treated as equivalent sources of kWh, although each has different limits, costs and wear.	A shared energy balance that enforces asset-specific constraints, minimum loading and reserve requirements.
Fragmented operational data	Equipment ratings, tariffs, operating records and load data live in different files and formats, with inconsistent units, timestamps and gaps.	Data onboarding with unit checks, gap reporting, provenance records and a reproducible project record.
Limited modelling and decision-support capacity	Smaller operators have no energy-modelling team; planning is done in spreadsheets and rules of thumb, and options are rarely compared.	A guided workflow where a non-modeller can load data, pick a strategy, and read a comparison with its assumptions.

2.4 Target users and stakeholders

The platform separates three roles. All three are hypotheses derived from secondary research and engineering observation; structured interviews with representative operators have not yet been completed, and that is a Phase 1 objective rather than a claim.

Role	Who they are	What they need from the product
DAILY USERS Site energy managers, operations engineers and facility operators	Electrical or operational training, but no modelling team.	A next-day view, two or three feasible plans, and a plain explanation of why one is preferred.
BUYERS Microgrid owners, utilities, energy-service companies and distributed infrastructure operators	Approving spend, carrying operating cost, answering to a board or a lender.	Operating-cost and reliability impact, payback logic, and evidence that the recommendation is defensible.
IMPLEMENTATION PARTNERS Engineering integrators, electrical engineering firms and equipment vendors	Supporting the site; confirming ratings and interconnection assumptions.	A common model to compare configurations; confirmed ratings and operating limits; data interfaces.

> **IMPORTANT QUALIFICATION**
>
> These user groups represent initial customer hypotheses. Their priorities, purchasing processes and willingness to pay must be tested through structured discovery and pilot engagement. No interview, letter of intent or procurement signal exists today.

Current-state operator journey (hypothesis). The sequence below describes how a site without islandOS decides today. It is reconstructed from secondary sources and engineering reasoning, not from recorded interviews.

Step	What happens now	Why it produces a decision gap
1	Read the current state: battery state of charge, PV output, load, grid or generator status.	Only the present moment is visible; nothing about the next operating period.
2	Set dispatch by experience and static rules: generator set-points, pumping or desalination windows.	Rules are tuned for typical days and become expensive on atypical ones.
3	No ahead-of-time preview under a different weather or demand assumption.	Alternatives cannot be compared before money is spent on fuel or imports.
4	React after a shortfall appears: start backup, shed flexible load.	Response is later and more expensive than a planned shift.
5	Limited comparison and audit: no systematic side-by-side runs, no record of assumptions.	Decisions cannot be reviewed, taught or defended afterwards.

2.5 Market context · 2.6 Competitive position

A real infrastructure need, and an honest account of where islandOS sits

Reliable distributed electricity is a live infrastructure need, not a niche. The World Bank and ESMAP's 2022 mini-grid market research identified 21,557 installed mini-grids and 29,353 planned projects across 138 countries and territories, serving an estimated 47.9 million people [1]. The World Bank separately reports that about 560 million people in Sub-Saharan Africa lacked electricity access in 2024 [2]. IRENA's 2025 grid-modernisation guidance for small island developing states treats storage, advanced energy management, forecasting and demand-side management as standard measures for resilient island systems [3].

> **WHAT THESE FIGURES DO AND DO NOT MEAN**
>
> The scale of distributed energy infrastructure indicates a relevant problem context, but it does not establish the addressable market for islandOS. The team's initial commercial research will focus on identifying sites where the value of better operational decisions can justify the cost of software adaptation, integration and ongoing use. Serviceable market has to be established site by site, through compatible facility types, equipment configurations, data availability, procurement requirements and willingness to pay. No TAM/SAM/SOM estimate is presented because the team cannot support one.

2.6 Current research, evidence and competitive position

The team has completed preliminary secondary research on remote mini-grids, distributed renewables, energy-management systems, forecasting and multi-resource optimisation. That research supports the framing above; it does not constitute customer validation.

Existing commercial EMS and SCADA platforms already provide forecasting and optimised dispatch. islandOS does not claim those platforms are incapable of these functions, nor that it is cheaper, better or more accurate. The question it explores is narrower and more defensible: whether a modular, simulation-first workflow with explicit assumptions can serve smaller operators who are blocked less by algorithms than by data integration, implementation complexity and the absence of specialist staff.

Alternative today	Strength	Where it falls short for the target user
Manual spreadsheets and rule-based dispatch	Low cost, familiar, no procurement friction.	Depends on individual experience; outputs are hard to audit or compare; no next-day preview.
Commercial EMS / SCADA platforms	Powerful, field-proven, integrated with control.	Implementation cost and integration effort may be difficult for a small operator to justify; the model is usually vendor-specific and closed.
External consultants and energy-service providers	Deep expertise, credible to financiers.	Periodic and project-shaped; rarely continuous operational decision support; recommendations often not reproducible by the site.
islandOS (positioning hypothesis)	Modular, simulation-first, transparent, reproducible, human-reviewed; runs on a laptop or a hosted demo.	Unvalidated: no operator interviews, no pilot, no field accuracy, no verified savings. The hypothesis, not the advantage, is what this phase tests.

Where the proposed differentiation would lie, if it survives validation:

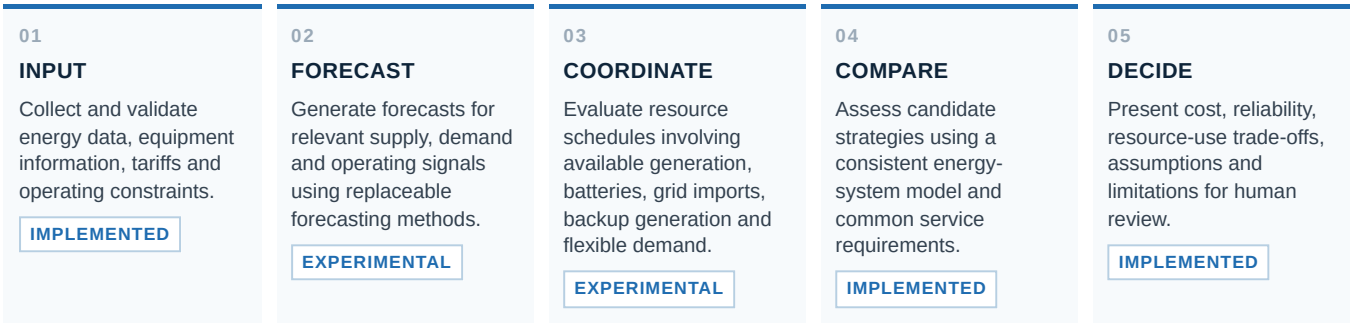
- Accessible data onboarding — the exports a small site already keeps, not a new telemetry stack.
- Explicit operating constraints — equipment limits and critical service stated, not weighted away.
- Forecast-informed scenario evaluation — the same physical model applied to each candidate strategy.
- Consistent energy and cost comparison — one accounting basis, one tariff basis, side by side.
- Reproducible decision records — assumptions, inputs, model version and outcome preserved.
- Human-reviewed recommendations — advisory only, outside the control path.

These characteristics are design objectives and positioning hypotheses, not yet independently validated competitive advantages. The team does not claim that islandOS is more advanced than existing commercial energy-management systems, and does not claim cost or accuracy superiority.

3. Proposed Idea — Overview

3.1 The five-step decision workflow

islandOS sits alongside existing infrastructure as a decision-support layer. It does not replace generation equipment, batteries, SCADA, protection systems or generator controllers. It uses the operating data and equipment information a site already has, evaluates possible conditions, compares alternatives, and presents recommendations for human review. Figure 2 shows the intended workflow together with its current maturity; it does not imply that all five stages are integrated in the hosted application.



Why the third stage is called Coordinate rather than Optimise. The current system does not guarantee global optimality in every scenario. It evaluates how available resources and flexible loads can operate together under declared constraints, and compares the resulting candidates. Calling that stage “coordination” keeps the claim aligned with the evidence.

Operational outcomes would return to INPUT — a long-term feedback loop, not yet closed in the hosted application. Each stage is a bounded capability with its own evidence trail; no stage may silently substitute for another.

3.2 Workflow mapping to the current implementation

The table connects each stage to the code that supports it today, so a reviewer can check the implementation rather than take the narrative at face value. Repository paths refer to the public default branch; experimental work lives in separate modules and is not part of the hosted end-to-end workflow.

Stage	Produces	Repository foundation	Current maturity
INPUT	Validated site record, asset inventory, assumption and provenance log	islandos_core/signals.py, configuration.py, dataset_provenance.py, data_quality.py	IMPLEMENTED — CSV/XLSX onboarding for supported schemas; universal site mapping not implemented.
FORECAST	Forecast series, baseline comparison, uncertainty metadata	islandos_core/forecasting.py, scenario_forecasting.py; experiments/	EXPERIMENTAL — statistical baselines with backtesting; no site-calibrated service.
COORDINATE	Candidate schedules, cost and reliability per plan	islandos_core/microgrid_model.py, schedule_evaluator.py; optimization/	EXPERIMENTAL — bounded scheduling search; no global-optimality claim.
COMPARE	Side-by-side strategy evaluation under one energy balance	islandos_core/schedule_evaluator.py, audit.py	IMPLEMENTED — within the simplified deterministic simulation scope.
DECIDE	Recommendation, rationale, cost comparison, audit record	app.py, islandos_core/decision.py, audit.py, islandos_advisor.py	IMPLEMENTED — comparison and audit foundations; economics as a screening module.

Rows labelled implemented can be checked against the default branch today; experimental rows exist as separate modules and are not wired into the hosted workflow. Where a stage carries two labels, the capability exists but its production wiring does not.

3.3 System Architecture

One core, several interfaces

The software separates core energy-system logic from the user interface, so one set of models can serve different application surfaces — the public demonstration, a customer console and an integration API — without duplicating the mathematics. Figure 3 shows the layering, including the capabilities that sit outside the current implementation.

INTERFACE LAYER
Public Streamlit demonstration · React + TypeScript local console foundation · Loopback reference API · CSV / Excel export

DECISION AND ECONOMICS LAYER
Strategy comparison · operating-cost screening · NPV and payback analysis · decision explanation · audit and evidence records · fail-closed governance gate

RESOURCE COORDINATION LAYER EXPERIMENTAL / NOT FULLY INTEGRATED
Shared energy-balance simulation · battery and generator constraints · grid import evaluation · flexible-load scheduling · scenario and sensitivity evaluation

FORECASTING LAYER EXPERIMENTAL
Statistical forecasting baselines · forecast candidate evaluation · chronological backtesting · forecast error analysis · experimental machine-learning candidates

DATA FOUNDATION
Data ingestion · data quality · asset and configuration registry · data provenance · project records · local SQLite store

Figure 3. Layered architecture. Core models are shared across interfaces; the decision and economics layer records what was assumed and what was recommended. Experimental status is marked inside the figure.

NOT PRESENT TODAY: live SCADA / PLC integration · equipment control commands · production cloud deployment · site-calibrated models · independent engineering certification. These are not implied by any other section of this document.

Module	Main responsibility
Data ingestion and provenance	Import and inspect time-series data; check units, timestamps, missing values and sources; preserve a record of what was supplied.
Asset and configuration registry	Record equipment ratings, efficiencies, operating limits and site configuration for reuse across scenarios.
Statistical forecasting baselines EXPERIMENTAL	Baselines for generation, load and price signals; chronological backtesting and error analysis.
Shared energy-balance simulation	Calculate energy flows, storage state and resource inventory under one shared accounting model.
Flexible-load scheduling EXPERIMENTAL	Exploratory scheduling of battery, diesel, grid import and flexible demand under declared constraints.
Strategy comparison and economics	Compare outcomes across conditions and uncertain assumptions; screen lifecycle cash flow, NPV, payback and sensitivity from user-entered inputs.
Decision interface and audit workspace	Present schedules, metrics, trade-offs and plain-language explanations; preserve configuration, inputs, run records and model version for later reproduction.

FIVE DESIGN PRINCIPLES THE ARCHITECTURE ENCODES

A shared physical simulation prevents each plan being scored on its own optimistic assumptions. A unified multi-resource view prevents separately optimising the battery and the diesel schedule. Explicit hard constraints prevent critical-service requirements being outvoted by a cost term. Reproducible records prevent a recommendation nobody can re-run. Human review prevents drift from advisory software into uncontrolled actuation.

3.4 Implementation Maturity · 3.5 Capability Status

Three layers, stated explicitly

Proposals commonly blur what is built with what is hoped for. This document separates three layers: code that runs today, capabilities that exist but are incomplete, and capabilities that require field work and approval before they can be offered to a customer.

Layer	What it contains	What is missing before a customer could rely on it
1 · IMPLEMENTED SOFTWARE FOUNDATIONS runs today	Supported CSV/XLSX onboarding · data-quality and provenance functions · asset and configuration records · simplified energy-system simulation · deterministic strategy comparison · hourly audit outputs · customer-economics screening · decision and evidence-record foundations · local React + TypeScript console foundation · loopback reference API · local Shadow Mode evidence ledger and governance foundations	Site-specific calibration, operator acceptance testing, integration with a real data feed, and any form of engineering certification.
2 · EXPERIMENTAL AND INTEGRATION-INCOMPLETE CAPABILITIES	Multi-signal forecasting and chronological backtesting · optional Random Forest forecast candidate · forecast-led scheduling · flexible-load coordination · bounded multi-resource optimisation experiments · rolling-horizon workflows · read-only telemetry contracts and mock-source prototypes	Unified site-calibrated forecast pipeline; general uncertainty-aware coordinator; telemetry adapters for real protocols; end-to-end wiring into a single customer workflow.
3 · FUTURE CAPABILITIES REQUIRING VALIDATION AND APPROVAL not deliverable now	Site-calibrated forecasting · general uncertainty-aware coordination · live site telemetry · hardware and controller integration · supervised operation · autonomous control, only if separately justified and approved	Site integration, cybersecurity architecture, electrical-safety case, qualified engineering review, explicit site authorisation and a separate regulatory path.

WHY THE LAYERING MATTERS FOR REVIEWERS

A reviewer can check layer 1 against the repository. Layer 2 is visible as experimental code but carries no customer promise. Layer 3 is the ambition, deliberately separated so that no part of this proposal can be read as a claim of deployed control. Any forward plan in Section 5 or Section 8 moves one layer at a time, with a stated gate between layers.

3.5 Capability status

Capability	Status	Evidence or boundary
Energy-system representation (PV, wind, battery, grid, diesel, essential and flexible load)	IMPLEMENTED	Within the documented simplified model scope; not a general digital twin of an arbitrary plant.
Deterministic energy accounting baseline	IMPLEMENTED	Common basis for comparing strategies; transparent priority policy.
Strategy comparison with hourly audit and CSV export	IMPLEMENTED	Baselines and audit outputs in the public app.
CSV / XLSX onboarding and selected import workflows	IMPLEMENTED	Validation and inspection foundations for supported schemas; not a universal connector.
Customer-economics screening (NPV, payback, break-even, sensitivity)	IMPLEMENTED	Computed from user-entered assumptions; results are scenarios, not verified customer savings.
Decision record, audit workspace, fail-closed governance gate	IMPLEMENTED	Records decisions and refusals with inputs and model version.
Multi-signal forecasting and backtesting	EXPERIMENTAL	Statistical baselines on synthetic or supplied data; not a trained, site-validated service.
Forecast-led constrained scheduling and rolling horizon	EXPERIMENTAL	Exploratory, bounded scopes; not part of the hosted end-to-end workflow.
Multi-objective search experiments	EXPERIMENTAL	Candidate comparisons under declared assumptions; no global-optimality claim.
Read-only telemetry and integration prototypes	EXPERIMENTAL	Contract and mock-source prototypes only; no live SCADA or PLC feed.
Site-calibrated multi-signal forecast service	PLANNED	Target capability; forecast uncertainty integration not delivered.
Live device connection, remote control, closed loop, autonomous control	LONG-TERM	Requires site authorisation, electrical-safety review, cybersecurity architecture and separate approval. No actuation path exists.
Verified savings, emissions reduction, battery-life improvement, ROI	NOT VALIDATED	No site data exists; only modelled potential under synthetic assumptions.

3.6 Public Demonstration · 3.7 Roadmap

What a visitor can actually do today, and the gated path forward

The public Streamlit prototype (islandos.streamlit.app) lets a visitor explore the built-in synthetic scenario or upload hourly data, adjust selected equipment and storage assumptions, compare operating strategies, inspect hourly outcomes, and export results. A separate rolling-horizon page hosts an experimental workflow.

> **WHAT THE HOSTED DEMONSTRATION IS, AND IS NOT**
>
> The public demonstration exposes selected planning and simulation workflows using synthetic data. The experimental forecasting and coordination workflows are not automatically available as one fully integrated hosted product. Users should not upload confidential operational data to the public demonstration; uploaded data is analysed as supplied and is not checked against a real site. The prototype does not connect to live SCADA, read live sensors, or send control commands to equipment.

3.7 Maturity roadmap — five gated stages

Each stage has an entry condition, an evidence requirement and an explicit gate before the next stage. Nothing advances because a milestone date arrives; it advances because evidence exists.

Stage	Content	Gate to the next stage
STAGE 1 — SOFTWARE FOUNDATION current position	Prototype, data workflows, simulation, strategy comparison and audit foundations.	A reviewer can reproduce a run from recorded configuration.
STAGE 2 — FORECASTING AND COORDINATION INTEGRATION	Improve forecast evaluation, connect experimental forecasting and coordination workflows, and test against consistent baselines.	Forecast and coordination methods evaluated against a common baseline on the same data.
STAGE 3 — SITE-SPECIFIC VALIDATION	Obtain authorised historical data, confirm equipment constraints with qualified engineers, compare against the operator's documented baseline.	Simulated balances fall within a documented tolerance of recorded history; constraints confirmed rather than assumed.
STAGE 4 — READ-ONLY SHADOW EVALUATION	Run alongside normal operations without issuing control commands; collect operator feedback and record recommendation acceptance or rejection.	Operators consider the recommendations credible enough to discuss a supervised trial.
STAGE 5 — CONTROLLED INTEGRATION AND SCALING	Only after separate engineering, cybersecurity, safety and site approvals, explore supervised integration and reusable deployment workflows.	Requires approvals this proposal does not request and does not assume.

The project advances through evidence gates, not simply by reaching a planned date. The team is currently at Stage 1 and is preparing the entry conditions for Stage 2 — a pilot partner, authorised historical data, and qualified engineering review of the site model. Stages 3–5 are deliberately placed outside this proposal's claims.

4. Innovation and Solution Design

4.1 Core innovation

Microgrid forecasting, physical simulation and optimisation are established fields. islandOS does not claim to invent forecasting, microgrid simulation or optimisation, and does not claim to outperform every commercial energy-management system. Its proposed contribution is to bring these functions into a coherent, transparent and accessible decision-support workflow.

The platform is designed around five principles:

Design principle	What it means in practice
A common system representation	Candidate strategies are evaluated against the same declared energy-balance model, rather than each plan being scored on its own optimistic assumptions.
Forecast-informed decisions	Changing supply, demand and operating signals can inform the evaluation of future operating conditions — the forecast is carried into the same model, not parked in a separate dashboard.
Constraint-aware coordination	Equipment limits and essential-service requirements are represented explicitly, so a cost term cannot quietly outvote a critical-load requirement.
Economic comparison	Alternatives are assessed not only by energy flows but also by operating cost and investment assumptions, on one tariff basis.
Reproducible reasoning	Assumptions, inputs, model versions and results are recorded so decisions can be reviewed, re-run and contested.

> **WHY THIS MATTERS**
>
> A forecast alone does not tell an operator what to do. An optimisation result alone does not tell a customer whether the investment is worthwhile. islandOS aims to connect prediction, operational constraints and economic consequences in one reviewable decision process.

What “AI-assisted” means here, and what it does not. The value proposition is not a single algorithm. It is the integration of data preparation, forecasting, resource coordination, economic evaluation and explainable human decision-making. AI-assisted refers primarily to data-driven forecasting and algorithmic decision-support methods. It does not mean that a large language model controls the energy system. Language-model assistance may support user interaction, but it never substitutes for a validated energy balance, an enforced operating limit or engineering review.

The current forecasting capabilities remain experimental and require independent site-specific evaluation. No accuracy figure is claimed.

> **FOUR CONDITIONS A RECOMMENDATION MUST SATISFY BEFORE AN OPERATOR WILL ACT ON IT**
>
> **Feasible** — it respects every confirmed equipment limit and service requirement. **Explainable** — it states which inputs and assumptions produced it, in language a site engineer accepts. **Comparable** — it is shown against at least one alternative and against the operator's current practice. **Reproducible** — it can be re-run from a recorded configuration. A recommendation missing any of the four is a demo, not a decision aid.

What islandOS is not. It is not a chatbot layered over energy data, not a vendor-specific digital twin, not an EMS or SCADA replacement, and not a forecast-accuracy claim. The analytical value sits in the physical model, the constraint set and the priced comparison.

4.2 Interdisciplinary Approach · 4.3 Limitations

Four disciplines, and the boundaries the team chooses to state

The project combines four academic disciplines, with complementary engineering perspectives across energy systems, water treatment, software and electrical infrastructure. Their integration is central to the design:

Discipline	What it contributes to islandOS
Nuclear / energy systems engineering	Energy-system analysis helps define system-level constraints and reliability requirements, and the habit of stating operating limits explicitly rather than burying them in a weighted objective.
Applied chemistry	Applied chemistry contributes process understanding for electricity-consuming treatment applications such as reverse osmosis: recovery ratio, specific energy demand, water-quality constraints and the coupling between water storage and pump scheduling.
Software engineering	Software engineering supports modular implementation, data quality, testing and reproducibility — the interfaces that let one core model serve a demo, a console and an API.
Electrical engineering (power systems)	Electrical engineering contributes power-system reasoning, equipment constraints and practical operating considerations — distribution-level behaviour, generator and battery limits, load characteristics, and the constraints a site engineer would refuse to violate.

This matters practically: an analytical result is only as useful as its input data and equipment assumptions. A model that schedules a pump at an hour the tank cannot accept it, or that treats a diesel generator as infinitely modulating, produces a plan no operator will follow.

4.3 Current technical limitations

The current implementation does not establish, and this proposal does not claim:

- No global optimality claim — the coordination stage evaluates candidates under declared assumptions.
- No site-specific forecast accuracy claim — statistical baselines only, evaluated on synthetic or supplied data.
- No verified battery lifetime improvement — throughput is reported as an operational indicator, nothing more.
- No verified emissions reduction or fuel saving — only modelled potential exists.
- No live device control and no live SCADA/PLC connection — there is no actuation path in the application.
- No independent safety certification and no interconnection or protection-coordination approval.

From advisory tool to control system — what must be proven	Current position
Constraint validity	Constraint set is declared and checked against equipment records; it is not yet reviewed by a qualified engineer for a specific site.
Command path and fail-safe behaviour	No actuation path exists. Any future path needs a defined fallback, interlocks and a documented failure mode.
Cybersecurity and data governance	Read-only, minimised data handling is the current design assumption; no site security assessment has been performed.
Authorisation and liability	Operational approval remains with site-authorised personnel; no certification, insurance or regulatory filing exists.

This is the boundary the team chooses to state rather than blur: islandOS is built to make a human decision better, and it will stay out of the control path until the four items above are proven for a specific site.

4.4 Development history

islandOS has a development history. An earlier prototype in the same team's development path addressed a narrower renewable-energy and water-production scheduling problem. Building on that foundation, the team extended the system into a general multi-resource decision-support platform: the single-process scheduling scope became a shared multi-resource energy-system framework with an asset and configuration registry; application-specific operating constraints became configurable site constraints; and customer-economics screening was added, which the earlier prototype did not contain. The water-and-energy application framing is retained as one use case, not as the boundary of the product.

5. Implementation Plan

5.1 Validation sequence

Months below are relative to project start or pilot approval, not fixed calendar dates. Each stage has an entry condition and an evidence requirement; a stage that fails its gate is revised rather than declared successful. The team does not commit to securing a pilot partner, and the plan describes what would be done with one rather than assuming one exists.

Stage	Activity	Deliverable and stop condition
1 · Stakeholder discovery months 1–2	Structured interviews with daily users and buyers; map existing data sources, meter availability, dispatch practice, critical loads, tariffs, fuel logistics and data permissions.	A bounded use case and a written interview record. Stop condition: no site willing to share authorised data in read-only form.
2 · Data availability and permission review months 1–2	Confirm meter coverage, sampling interval, export format, gaps, ownership, redaction requirements and permitted use; complete cybersecurity and privacy review before any operational data is handled.	A data-permission record and a redaction plan. Stop condition: data too sparse to support a credible model.
3 · Site configuration and engineering review months 2–4	Map meters and equipment to model components; confirm ratings, efficiencies and operating limits with a qualified engineer; record tariff and fuel assumptions; compare simulated balances with history.	A site-specific configuration signed off by a qualified site engineer. Gate: simulated balances within a documented tolerance of recorded history.
4 · Retrospective comparison months 4–6	Replay historical periods on identical input data; examine grid purchases, diesel use, renewable utilisation, curtailment, critical-load service and battery throughput; check whether the recommended schedule was physically feasible at the time.	A comparison report the site can audit. Gate: feasible schedules, and at least one agreed objective improves without degrading critical-load service.
5 · Read-only shadow evaluation months 6–9	Run alongside normal operations without issuing any control command; log every disagreement between the recommendation and the operator's decision, with reasons captured in writing.	A shadow-trial log and an operator feedback summary. Gate: rejection reasons are understood rather than dismissed.
6 · Controlled iteration months 9–12	Data adapters, model refinement, reporting and usability work driven by pilot findings, not internal preference.	A revised platform with a change log tied to pilot findings.
7 · Separate approval for any operational integration	Not requested by this proposal. Requires engineering, cybersecurity, safety and site approvals.	No supervised or closed-loop operation without those approvals.

5.2 Alignment with the GSC phases

Phase	What it is for, and where islandOS stands
Phase 1 — IGNITE current submission	Establish a well-defined problem, a credible solution concept, a clear prototype boundary and a realistic validation plan. This document is that submission.
Phase 2 — BUILD proposed	Focus on development, testing, iteration and user feedback. The team intends to use these activities to determine whether the proposed decision-support workflow can produce useful results under realistic operating conditions. Phase 2 is not claimed as completed.
Phase 3 — SHINE proposed	Present the refined prototype, evidence and a realistic implementation pathway. Phase 3 is not claimed as completed, and no evidence beyond Stage 1 exists today.

5.3 Required resources

> **RESOURCES THE PILOT WOULD NEED**
>
> A pilot site operator for authorised historical and shadow-mode data · a qualified electrical engineer or integrator to confirm ratings and sign off the site model · a site operations contact to judge whether a plan is realistic · team software capacity for adapters, model refinement and reporting · a cybersecurity and privacy review before sensitive data is handled, even read-only.

The initial pilot should avoid purchasing new generation, storage or process equipment. The first question is whether better decisions on existing assets create value before any capital is spent.

6. Customer Economics and Business Logic

A technically feasible recommendation is not automatically a commercially valuable recommendation

islandOS must demonstrate that the value of improved operating decisions can exceed the cost of software adaptation, integration, maintenance and user training. The platform should also be able to identify situations where adoption is not economically justified — stating the negative case is part of the product, not an admission against it.

Verified operating benefit
− software and service cost
− integration cost
− maintenance and training cost
= NET CUSTOMER VALUE

This framework is proposed for future site validation. **No numerical customer benefit has been established.** No savings percentage, payback period or subscription price is claimed today.

What the buyer is actually trying to answer is narrower than a model output: does this lower energy operating cost; does it reduce avoidable diesel use; does it use available renewable generation better; does it avoid unnecessary grid purchases; does it keep critical loads served; and is it worth buying, deploying and maintaining at all?

Measure	Definition
Gross operating benefit	Baseline operating cost minus optimised operating cost, over the same period, with the same service requirements.
Net customer value	Gross operating benefit minus software, integration and maintenance cost.
Cost components	Grid electricity charges, diesel fuel expenditure, generator variable O&M, and a battery-degradation allowance only where a defensible estimate exists.
Reliability side-conditions	Unmet critical energy, unmet water demand and reserve violations are reported, never netted out against cost.
Sensitivity	Results re-run across tariff, fuel-price, resource-availability and forecast-error assumptions; a saving that exists only at one assumption is not a saving.

Illustrative worked case, no measured figures. Consider an operator deciding whether to shift a flexible load's operating window. On the same physical model and evaluation period, islandOS compares the baseline strategy's grid electricity and fuel cost against the candidate strategy's; it reports battery throughput, whether essential loads and water supply remained served, and the software integration cost. The comparison states whether the adjustment yields a positive net customer value and under which assumptions. No saving amount is assumed here; any figure would have to come from a real site baseline.

> **WHEN ADOPTION WOULD BE WORTHWHILE — AND WHEN IT WOULD NOT**
>
> Worthwhile only after a baseline is established: gross operating benefit covers software, integration and maintenance cost; critical-load service is not worsened; and the improvement holds under sensitivity. Not worthwhile where modelled savings do not cover those costs, where data are too poor to support a credible model, where simple manual rules already meet the site's needs, or where adopting the software would introduce unacceptable service risk.

Pilot budget position. The current prototype was built with open-source software and personal computing resources. Cost categories in a future pilot budget would include software engineering and site-specific model adaptation, data mapping and integration, qualified engineering review, site visits and training, computing and storage, and cybersecurity, privacy and legal review. A reliable budget cannot be quoted until a pilot site, data interface, staffing rate and security requirement are agreed. The pilot budget deliberately excludes new generation, storage or process equipment, control hardware and retrofit labour, permanent headcount, and certification or independent audit.

7. Impact

Four intended directions, each subject to validation

If validated, islandOS would help operators use available renewable electricity better, reduce avoidable diesel consumption and unnecessary grid purchases, coordinate flexible demand with available supply, maintain appropriate reserves, and make operating decisions more transparent. The platform does not generate electricity and does not guarantee lower emissions.

Intended impact — subject to validation	What would have to be measured
Better renewable energy utilisation	Renewable energy used and curtailed, on the same period as the baseline, from meter data.
Reduced avoidable diesel and grid expenditure	Litres consumed, generator running hours and metered grid charges, from logs and invoices.
More reliable coordination of essential and flexible demand	Unmet critical energy, outage hours and reserve violations, reported separately from cost.
More transparent and accountable energy decisions	Recommendation acceptance and rejection reasons, captured in the shadow trial.

> **A HARD PRIORITY**
>
> Critical-service reliability must never be sacrificed simply to produce a lower modelled operating cost. A cost reduction reported while service has degraded is not a gain.

7.1 Impact measurement

The project distinguishes modelled results from measured results. A simulated reduction is not a real-world saving until it is validated against meters, invoices, generator records or other site evidence.

Impact area	Indicator	Verification method
Renewable utilisation	Renewable energy used; curtailment; grid imports	Meter data and energy-balance records over the baseline period.
Fuel consumption	Diesel consumed; generator running hours	Generator logs and fuel purchase records.
Grid expenditure	Grid charges; operating cost	Actual tariffs and invoices — not modelled prices.
Critical-load reliability	Unmet critical energy; outage hours; interruptions	Site logs and operational records, reported separately from cost.
Battery throughput	Charge/discharge throughput; equivalent cycles	Battery management system records where available. An operational indicator only — it does not prove longer battery life.
Operator acceptance	Recommendation acceptance; rejection reasons; planning time	Shadow-trial records and structured feedback.

7.2 Verification stages and success conditions

Evaluation proceeds in three stages: retrospective comparison against the historical baseline on identical data; read-only shadow evaluation alongside normal operations; and, only later and separately authorised, a supervised operational evaluation requiring site authorisation, engineering review and safety approval. Every comparison preserves input data, configuration, tariff assumptions, model version and scenario definitions.

- Schedules satisfy confirmed physical and operating constraints.
- Critical-load service is not worsened against the baseline.
- The operator can understand the assumptions and trade-offs without modelling training.
- At least one agreed objective improves without unacceptable deterioration elsewhere.
- The comparison can be reproduced by someone outside the team from recorded configuration.

Social and equity considerations. A decision-support tool can exclude the people it is meant to serve in three ways: by requiring skills the site does not have, by requiring data the site cannot produce, and by encoding priorities the community never agreed to. Outputs are written in plain language and paired with the assumption set; the workflow accepts the CSV and Excel exports small sites already keep; critical loads are entered as explicit constraints by the site rather than inferred from historical consumption that may already have under-served some users. None of this is yet tested with a real community, and involving operators and affected users in defining critical loads is a stated pillar of the first pilot rather than an achieved outcome.

The project will evaluate sustainability through measurable operational indicators rather than unsupported claims of environmental benefit. No emissions reduction is claimed.

8. Scalability and Business Model

8.1 Technical scalability · 8.2 Geographic and use-case expansion

islandOS is designed around a common application core with replaceable, site-specific data adapters. The workflow can be configured for different combinations of solar, wind, storage, grid, diesel and flexible demand. A new site must not reuse another site's assumptions: each needs its own equipment records, tariffs, load characteristics, resource profiles and constraints. Scaling therefore means adding validated adapters and models, not multiplying an unvalidated claim.

The initial application domain is energy-constrained distributed infrastructure, where multiple energy resources and operating constraints must be coordinated. Potential future applications include:

- Island and remote microgrids
- Rural and weak-grid communities
- Distributed industrial facilities
- Commercial and institutional facilities with hybrid energy systems
- Critical infrastructure
- Energy-service company portfolios

Flexible demand may include water pumping, treatment processes, cold storage and selected industrial operations. Desalination and pumping are therefore one possible application among several, not the product boundary. Each new application requires its own data mapping, equipment representation, operating constraints and validation; the platform is designed for modular expansion, not assumed universal compatibility. Geographic expansion follows successful site-specific validation rather than preceding it.

8.3 Business model hypotheses

Three questions decide whether this is a business rather than a tool: who signs, what they are buying, and what makes the value visible. The buyer is normally the asset owner, the utility or the energy-service company that carries operating cost; the daily user is the site engineer who approves the plan. What they buy is foresight, comparison and documentation.

Revenue hypothesis	What the customer pays for	Status
Software subscription	Continuous access to forecasting, scenario comparison and decision records for one or more sites.	COMMERCIAL HYPOTHESIS — NOT YET VALIDATED. No pricing set, no customer signed.
Site assessment and configuration	A one-off engineering engagement that builds the site model, validates data and confirms constraints.	COMMERCIAL HYPOTHESIS — scope and price to be defined with a pilot partner.
Forecasting and coordination services	Recurring analytical service where the site prefers not to operate the model itself.	COMMERCIAL HYPOTHESIS — depends on validated accuracy and turnaround.
Integration and deployment services	Data-interface work with existing SCADA/EMS or metering, and training for operators.	COMMERCIAL HYPOTHESIS — requires cybersecurity and engineering review per site.
Future equipment and controller partnerships	A future bundled offering where a read-only or supervised interface is certified.	LONG-TERM — no partner approached, no integration certified.

Why not simply buy an existing energy-management tool? For a large utility-scale site, the honest answer is that it probably should. islandOS targets the segment where a full EMS implementation is hard to justify, where data lives in mixed formats, and where the operator needs a defensible comparison more than an automated controller. The entry point is advisory, which means it can be adopted without touching the control layer — the lowest-risk way to earn the evidence a larger commitment would require.

Regulatory and policy considerations. Future deployment must comply with local electrical-safety regulations, interconnection requirements, generator and battery operating rules, data-protection and cybersecurity policies, and site procurement and authorisation. The current software's export calculations do not establish that a schedule satisfies power-flow or interconnection requirements. Any live control integration would require a separate engineering safety case, a cybersecurity architecture, qualified review and explicit site authorisation.

9. Team and Execution

Five undergraduate students, four disciplines, two countries

islandOS is a collaborative project developed by five undergraduate students across four academic disciplines and two countries. The team jointly contributes to project conception, technical discussions, solution design, analysis, software development and iterative evaluation. Each member brings complementary academic knowledge, while technical decisions and implementation work are reviewed collaboratively. Roles below describe the main contribution directions agreed by the team; they overlap in practice and continue to evolve.

Member	Institution, programme, year	Contribution direction
Haochuan Lin	Shenzhen University BEng, Nuclear Engineering and Nuclear Technology Year 3	Project initiation and overall product vision; energy-system analysis; system architecture planning; technical coordination; core software development; system integration.
Liyue Cai	South China University of Technology BEng, Applied Chemistry Year 3	Water-treatment process analysis; reverse-osmosis process considerations; energy-demand constraints associated with treatment applications; interdisciplinary solution design.
Yunzhu Chen	South China University of Technology BEng, Software Engineering Year 3	Software engineering; software architecture; data processing; application development; code implementation; integration and testing.
Saya Parajuli	Tribhuvan University, Institute of Engineering, Pulchowk Campus BE, Electrical Engineering Year 4	Electrical engineering and power-system analysis; energy forecasting and load-coordination concepts; system design; software development; technical analysis.
Sonika Subedi	Agriculture and Forestry University BE, Electrical Engineering Year 4	Electrical engineering and distributed energy-system analysis; resource-coordination concepts; system design; software development; technical testing.

HOW THE TEAM ACTUALLY WORKS

The team combines energy-system reasoning, electrical engineering, water-treatment process knowledge and software engineering to develop a practical decision-support platform. Design questions are reviewed across disciplines before they are coded: the process side challenges whether a schedule is physically acceptable at the tank and membrane, the power side challenges whether it is acceptable at the bus and the generator, and the software side challenges whether it can be reproduced and audited. Every member contributes to conception, design and implementation review rather than to an isolated module.

Execution discipline	How it is applied
Traceable decisions	Design decisions and their reasons are recorded in the repository so a new member can reconstruct context instead of re-litigating it.
Automated checks	Tests, data-validation checks and packaging checks run in the public repository workflow; passing checks are evidence of software behaviour, not of field performance.
Honest scope control	New claims require code, an audit trail and a stated boundary; a feature is not described as available until it runs on the default branch.
AI use, reviewed	AI tools are used as drafting and analysis aids; each output is reviewed, modified and tested by a team member who remains accountable for it.

All five members are enrolled undergraduate students and are registered on the GSC team portal. **No professional engineering experience, laboratory resource, certification or industry tenure is claimed for any member.** Team credibility rests on transparent assumptions, auditable runs and clearly stated validation boundaries rather than on assumed field evidence.

10. Supporting Materials

10.1 Public repository · 10.2 Demonstration · 10.3 References

Repository: **github.com/HaochuanLin/islandOS**, default branch main. The public repository provides access to source code, documentation, tests and development history. The default branch contains the established software foundation, while experimental workflows and ongoing development may be maintained separately. Reviewers should refer to the current default branch when assessing implementation status.

The repository provides a transparent development record, including implementation foundations, experimental workflows, documentation and automated checks. The hosted demonstration provides a limited public view of selected planning and simulation functions. **Neither should be interpreted as evidence of commercial deployment or field validation.** File counts, line counts and commit history are a development record, not an innovation claim.

Technical area	Repository resources (default-branch paths)
Physical energy-system model	islandos_core/microgrid_model.py; tools/run_integrated_microgrid_demo.py
Schedule evaluation and comparison	islandos_core/schedule_evaluator.py
Decision and advice	islandos_core/decision.py; islandos_advisor.py
Audit, provenance, data quality	islandos_core/audit.py, dataset_provenance.py, data_quality.py
Forecasting (experimental)	islandos_core/forecasting.py; islandos_core/scenario_forecasting.py
Signals and configuration	islandos_core/signals.py; islandos_core/configuration.py
Coordination experiments	optimization/grid_search_optimizer.py; optimization/nsga2_dispatch_optimizer.py
Experimental rolling-horizon page	pages/2_Rolling_Horizon_Lab.py
Sensor and telemetry prototypes (experimental)	islandos_sensors/contract.py; islandos_sensors/mock_source.py
React + TypeScript console foundation	frontend/src/App.tsx; frontend/package.json
Shadow Mode evidence and governance	islandos_shadow.py; islandos_core/shadow_evidence.py; islandos_core/safety_governance.py
Continuous integration and tests	.github/workflows/python-unit-tests.yml; tests/
Design and evidence documentation	docs/PRODUCT_SYSTEM_BLUEPRINT.md; docs/ENGINEERING_MATURITY_ROADMAP.md

Public demonstration: islandos.streamlit.app — a prototype using synthetic data, not a deployed customer system. Users should not upload confidential or sensitive site data.

10.3 References

- [1] World Bank Group / ESMAP (2022). *Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers*, Table O.3. esmap.org
- [2] World Bank Group (2025). *Mission 300: An FAQ on Bringing Electricity to 300 Million in Africa.* worldbank.org
- [3] International Renewable Energy Agency (IRENA) (2025). *Powering Resilient Islands: Grid Modernisation Toolkit for SIDS.* irena.org
- [4] islandOS public repository. github.com/HaochuanLin/islandOS

Third-party code, assets and licences. The project uses open-source libraries and publicly documented engineering methods. Required copyright notices, licence texts and attribution statements must be retained in accordance with the applicable licences. The project does not claim established engineering methods, third-party libraries or externally authored work as original inventions. The repository carries an MIT licence file and dependency manifests; a separate NOTICE file is not yet present, so attribution obligations are tracked in the repository rather than asserted here.

Appendix A — Claim-Verification Table

Status labels follow the definitions on page 2; a merged pull request is not field validation. Statuses here align with Sections 1, 3.1, 3.3, 3.4 and 3.5.

Claim in this proposal	Evidence	Status	Qualification
Public interactive strategy screening on synthetic data	Hosted Streamlit app; evaluator and audit modules	IMPLEMENTED	Demo uses synthetic data; not a deployed customer system.
Simplified deterministic energy-system simulation	islandos_core/microgrid_model.py	IMPLEMENTED	Simplified physical model; not a general digital twin or site-validated model.
Supported CSV/XLSX onboarding with provenance	signals.py, data_quality.py, dataset_provenance.py	IMPLEMENTED	Supported schemas only; not a universal site connector.
Customer-economics screening (cash flow, NPV, payback, break-even, sensitivity)	Economics module; console screens	IMPLEMENTED	Computed from user-entered assumptions; not verified savings.
Decision record, audit workspace and fail-closed governance gate	Decision, advisory and audit modules; SQLite evidence store	IMPLEMENTED	Records recommendations and refusals; not an operational control system.
Read-only Shadow Mode evidence ledger with digest and hash-chain checks	islandos_shadow.py; islandos_core/shadow_evidence.py	IMPLEMENTED	Local SQLite ledger; no live site has been observed.
Local React + TypeScript console foundation and loopback reference API	Console application; reference API	IMPLEMENTED	Local foundations, not a hosted production SaaS platform.
Experimental multi-signal forecasting with chronological backtesting	forecasting.py, scenario_forecasting.py, experiments/	EXPERIMENTAL	Statistical baselines; no site accuracy established; not a production forecast service.
Optional Random Forest forecast candidate	experimental model workflows	EXPERIMENTAL	One candidate among others; no reproducible comparison establishes superiority over other methods, and no data-leakage-free benchmark is claimed.
Forecast-led constrained scheduling and rolling horizon	Rolling-horizon page; scheduling experiments	EXPERIMENTAL	Exploratory scope; not part of the hosted end-to-end workflow.
Flexible-load coordination workflows	Background evaluation modules	EXPERIMENTAL	Bounded experiments; not wired into the hosted workflow.
Multi-objective search experiments (grid search, NSGA-II)	optimization/grid_search_optimizer.py, nsga2_dispatch_optimizer.py	EXPERIMENTAL	Candidate comparison under declared assumptions; no global-optimality claim.
Read-only telemetry contracts and mock sources	islandos_sensors/contract.py, mock_source.py	EXPERIMENTAL	Contracts and mock sources only; no live SCADA or PLC feed.
Site-calibrated forecast service	Documented gap items	PLANNED	Target capability; not delivered as a usable workflow.
Fully integrated customer decision workflow	Hosted app exposes selected modules only	PLANNED / INCOMPLETE	Stages exist separately; single end-to-end customer workflow not complete.
Live SCADA/PLC integration and remote control	None	LONG-TERM	Requires site integration, safety case, cybersecurity review, authorisation.
Autonomous equipment control	Outside prototype scope	NOT IMPLEMENTED	No actuation path exists in the application.
Verified savings, emissions reduction, battery-life improvement, ROI	None available	NOT VALIDATED	No site data; only modelled potential under synthetic assumptions.
Five-member joint development	Team records and contribution table	IMPLEMENTED	Complementary roles; no professional experience claimed beyond academic work.

REVIEWER CHECKLIST — FIVE MINUTES, NO ACCOUNT REQUIRED

1. Open the repository and confirm the default branch is main. 2. Open representative paths and read their scope notes: islandos_core/microgrid_model.py, islandos_core/forecasting.py, islandos_shadow.py, frontend/src/App.tsx and tests/. 3. Run the built-in synthetic scenario in the hosted demo. 4. Read every figure in this document as a design description, not as a measurement. Anything that cannot be checked in those four steps is labelled experimental, planned or not validated in the table above.

Final Statement

What the team has built, what it has not, and what it intends to prove

islandOS is an early-stage software and engineering project addressing the challenge of coordinating distributed energy resources under changing operating conditions. The team has developed a software foundation for data onboarding, simplified energy-system simulation, strategy comparison, customer-economics screening and auditable decision support. Experimental forecasting and resource-coordination workflows provide a basis for further integration and evaluation.

The next step is not to claim commercial readiness, but to establish whether the platform can deliver useful, feasible and economically defensible decisions in a real operating environment. Through authorised site data, qualified engineering review, retrospective evaluation and read-only operator feedback, the team aims to develop islandOS from a reproducible prototype into a practical decision-support platform. Its longer-term ambition is to support a wider range of energy-constrained sites and distributed infrastructure through modular data interfaces, replaceable analytical methods and site-specific validation.

Originality Statement

islandOS is a collaborative project conceived and developed by the five-member student team identified in Section 9. Its overall product direction, application framing, system architecture and ongoing implementation have evolved through the team's interdisciplinary discussions, engineering analysis, design decisions, software development and iterative evaluation. The project builds on established engineering principles, mathematical methods and open-source software components, and on an earlier prototype in the same team's development path acknowledged in Section 4.4. **The team does not claim the original invention of forecasting algorithms, optimisation algorithms, microgrid simulation or existing engineering methods;** its contribution lies in the product direction, system design, engineering implementation, integration, analysis and iteration. Relevant external sources and software licences are acknowledged where required.

AI-Assisted Work Declaration

The team used generative AI tools during different stages of project development and preparation of this submission. AI-generated outputs were reviewed, evaluated, modified and tested by team members before being incorporated. The team remains responsible for the accuracy, originality, technical integrity and final decisions presented in this document. AI tools were not used as substitutes for real-world validation, and no synthetic or illustrative data is presented as measured field evidence.

Area of AI assistance	How it was used	Human review applied
Research and technical analysis	Background organisation, summarisation, comparison of public sources.	Claims checked against cited public sources; content decisions and boundaries set by the team.
Brainstorming	Option generation for scope, framing and open questions.	Team selected, rejected and narrowed the options; final scope set by the team.
Software architecture and coding	Drafting code, explaining errors, suggesting refactors, architecture options and test cases.	Team reviewed, integrated, ran tests and modified each implementation before inclusion.
Debugging and documentation	Error diagnosis and first drafts of module notes.	Verified by running the code and reading the repository record.
Presentation and drafting	Diagrams, layout, structuring and language editing of this document.	Team selected, corrected and approved all figures, captions and factual statements.

Accuracy and Authenticity Declaration

To the best of the team's knowledge, the information in this proposal is accurate and complete. Technical claims reflect the public repository and hosted demonstration as of the submission date, with implementation maturity and limitations explicitly identified.

- All built-in demonstration data are synthetic.
- No verified customer savings are claimed.
- No field-validated forecast accuracy is claimed.
- No commercial deployment is claimed.
- No live equipment control is claimed.
- Experimental and planned functions are explicitly labelled throughout.

Submission: Global Sustainability Challenge 2026–27, Phase 1 (IGNITE), Sustainable Energy theme. Contact and repository details are given on the cover page.